

Q. NO. Define the convergence and the theorem:

$$\text{If } y_{t+2} + \frac{1}{6} y_{t+1} - \frac{1}{6} y_t = 2$$

① If

$$\begin{matrix} \Delta_1 \\ \Delta_2 \end{matrix} < 0 \quad (-ve)$$

$$\begin{cases} a_0 = 1 \\ a_1 = \frac{1}{6} \\ a_2 = -\frac{1}{6} \end{cases}$$

The case will be divergent

② If

$$\begin{matrix} \Delta_1 \\ \Delta_2 \end{matrix} > 0 \quad (+ve)$$

The case will be convergent

If $\Delta_1 > 0$ and $\Delta_2 < 0$ The case will point of inflection.

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} a_0 & 0 & a_2 & a_1 \\ a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix}$$

Example No 4

Does the time path of the equation

$$y_{t+2} + 3y_{t+1} + 2y_t = 12 \quad \text{Convergence?}$$

Solution

$$y_{t+2} + 3y_{t+1} + 2y_t = 12$$

here $a_0 = 1$ $a_1 = 3$ $a_2 = 2$

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3$$

$$\Delta_1 = -3$$

Now

$$\Delta_2 = \begin{vmatrix} a_0 & 0 & a_2 & a_1 \\ a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 2 & 3 \\ 3 & 1 & 0 & 2 \\ 2 & 0 & 1 & 3 \\ 3 & 2 & 0 & 1 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 3 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 0 & 1 \end{vmatrix} + 2 \begin{vmatrix} 3 & 1 & 2 \\ 2 & 0 & 3 \\ 3 & 2 & 1 \end{vmatrix}$$

$$-3 \begin{vmatrix} 3 & 1 & 0 \\ 2 & 0 & 1 \\ 3 & 2 & 0 \end{vmatrix}$$

$$\Delta_2 = 1 [1(1-0) - 0(0-6) + 2(0-2)] - 0 + 2 [3(0-6) - 1(2-3) + 1] - 3 [3(0-2) - 1(0-3) + 0]$$

$$\Delta_2 = 1 [1-4] - 0 + 2 [-18+7+8] - 3 [-6+3]$$

$$= -3 - 6 + 9 = -9 + 9 = 0$$

hence $\Delta_1 = -3 < 0$

$\Delta_2 = 0$

The time path is divergent Ans

Example: No. 5. - Test the convergence of the path of $y_{t+2} + \frac{1}{6} y_{t+1} - \frac{1}{6} y_t = 2$ by the Shur theorem.

Solution:

$$y_{t+2} + \frac{1}{6} y_{t+1} - \frac{1}{6} y_t = 2$$

here $a_0 = 1$ $a_1 = \frac{1}{6}$ $a_2 = -\frac{1}{6}$ $n = 2$

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & -\frac{1}{6} \\ -\frac{1}{6} & 1 \end{vmatrix} = 1 - \frac{1}{36}$$

$$\Delta_1 = 36 - \frac{1}{36} = \frac{35}{36}$$

Now we find Δ_2

$$\Delta_2 = \begin{vmatrix} a_0 & 0 & a_2 & a_1 \\ a_1 & a_0 & 0 & a_2 \\ a_2 & 0 & a_0 & a_1 \\ a_1 & a_2 & 0 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1/6 & 1/6 \\ 1/6 & 1 & 0 & -1/6 \\ -1/6 & 0 & 1 & 1/6 \\ 1/6 & -1/6 & 0 & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 1 & 0 & -1/6 \\ 0 & 1 & 1/6 \\ -1/6 & 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 1/6 & 0 & -1/6 \\ -1/6 & 0 & 1/6 \\ 1/6 & 0 & 1 \end{vmatrix} + (-1/6) \begin{vmatrix} 1/6 & 1 & -1/6 \\ -1/6 & 0 & 1/6 \\ 1/6 & -1/6 & 1 \end{vmatrix} - 1/6 \begin{vmatrix} 1/6 & 1 & -1/6 \\ -1/6 & 0 & 1/6 \\ 1/6 & -1/6 & 1 \end{vmatrix}$$

$$= 1 \left[1(1-0) - 0(0 + \frac{1}{36}) - \frac{1}{6}(0 + \frac{1}{6}) \right] - 0 \left[\frac{1}{6}(0 + \frac{1}{6}) - 1(-\frac{1}{6} - \frac{1}{36}) \right] - \frac{1}{6} \left(\frac{1}{36} - 0 \right) - \frac{1}{6} \left[\frac{1}{6}(0 + \frac{1}{6}) - 1(0 - \frac{1}{6}) + 0 \right]$$

$$\Delta_2 = 1 \left(1 - 0 - \frac{1}{36} \right) - \frac{1}{6} \left(\frac{1}{216} + \frac{7}{36} - \frac{1}{216} \right) - \frac{1}{6} \left(\frac{1}{36} + \frac{1}{6} \right)$$

$$= 1 \left[\frac{36-1}{36} \right] - \frac{1}{6} \left[\frac{1+42-1}{216} \right] - \frac{1}{6} \left[\frac{1+6}{36} \right]$$

$$\Delta_2 = \frac{35}{36} - \frac{42}{1296} - \frac{7}{216} = \frac{1260-42-42}{1296} = \frac{1176}{1296}$$

$$\Delta_1 = \frac{1176}{1296}$$

Hence $\Delta_1 = 35/36 > 0$

$$\Delta_2 = 1176/1296 > 0$$

The time path is convergent Ans.

Exercise: No. 18.4

Q: No. 6. Test the convergence of the solutions of the following difference equations by the Schur theorem.

(a) $y_{t+2} + \frac{1}{2}y_{t+1} - \frac{1}{2}y_t = 3$

Solution: here $a_0 = 1$, $a_1 = \frac{1}{2}$, $a_2 = -\frac{1}{2}$

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -1/2 \\ -1/2 & 1 \end{vmatrix} = 1 - \frac{1}{4} = \frac{4-1}{4}$$

$$\Delta_1 = 3/4$$

Now we find the Δ^2

$$\begin{vmatrix} 1 & 0 & 0 & a_1 \\ a_1 & 1 & 0 & a_2 \\ a_2 & 0 & 1 & a_1 \\ a_1 & 0 & a_1 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -1/2 & 1/2 \\ 1/2 & 1 & 0 & -1/2 \\ -1/2 & 0 & 1 & 1/2 \\ 1/2 & -1/2 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 1/2 \\ 1/2 & 0 & 1 \end{vmatrix} \rightarrow \begin{vmatrix} 1/2 & 1 & -1/2 \\ -1/2 & 0 & 1/2 \\ 1/2 & -1/2 & 1 \end{vmatrix} \rightarrow \begin{vmatrix} 1/2 & 1 & 0 \\ -1/2 & 0 & 1 \\ 1/2 & -1/2 & 1 \end{vmatrix}$$

$$= \left[1(1-0) - \frac{1}{2}(0 + \frac{1}{2}) \right] - \frac{1}{2} \left[\frac{1}{2}(\frac{1}{4}) - 1(-\frac{1}{2} - \frac{1}{4}) - \frac{1}{2}(\frac{1}{4}) \right]$$

$$- \frac{1}{2} \left[\frac{1}{2}(\frac{1}{2}) - 1(-\frac{1}{2}) \right]$$

$$= (1 - \frac{1}{4}) - \frac{1}{2}(\frac{1}{8} + \frac{1}{2} + \frac{1}{4} - \frac{1}{8}) - \frac{1}{2}(\frac{1}{4} + \frac{1}{2})$$

$$= \frac{3}{4} - \frac{1}{2} \left(\frac{1+4+2-1}{8} \right) - \frac{1}{2} \left(\frac{1+2}{4} \right)$$

$$= \frac{3}{4} - \frac{1}{2}(\frac{6}{8}) - \frac{1}{2}(\frac{3}{4}) = \frac{3}{4} - \frac{6}{16} - \frac{3}{8}$$

$$= \frac{12-6-6}{16} = \frac{12-12}{16} = \frac{0}{16} = 0$$

$$\Delta_2 = 0$$

$$\text{Since } \Delta_1 = \frac{3}{4} > 0$$

$$\Delta_2 = 0$$

Thus the time path is not convergent.

Ans

$$(b) \quad y_{t+2} - \frac{1}{9} y_t = 1$$

Solution: here $a_0 = 1$, $a_2 = -\frac{1}{9}$, $a_1 = 0$

$$\Delta_1 = \begin{vmatrix} a_0 & a_2 \\ a_2 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & -\frac{1}{9} \\ -\frac{1}{9} & 1 \end{vmatrix}$$

$$\Delta_1 = 1 - \frac{1}{81} = \frac{81-1}{81} = \frac{80}{81}$$

$$\Delta_1 = \frac{80}{81} > 0$$

Now

$$\Delta_2 = \begin{vmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_0 & a_2 & a_3 \\ a_2 & a_1 & a_0 & a_3 \\ a_3 & a_2 & a_1 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & -\frac{1}{9} & 0 \\ 0 & 1 & 0 & -\frac{1}{9} \\ -\frac{1}{9} & 0 & 1 & 0 \\ 0 & -\frac{1}{9} & 0 & 1 \end{vmatrix}$$

$$\Delta_2 = \begin{vmatrix} 1 & 0 & -\frac{1}{9} \\ 0 & 1 & 0 \\ -\frac{1}{9} & 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 & -\frac{1}{9} \\ -\frac{1}{9} & 0 & 0 \\ 0 & -\frac{1}{9} & 1 \end{vmatrix} - 0$$

$$= 1 \left[1(1-0) - 0 - \frac{1}{9}(0+\frac{1}{9}) \right] - \frac{1}{9} \left[0(0+0) - 1(-\frac{1}{9}) - \frac{1}{9}(\frac{1}{9}) \right]$$

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$$1 - \left(1 - \frac{1}{81}\right) - \frac{1}{19} \left(\frac{1}{9} - \frac{1}{729}\right)$$

$$= \frac{81-1}{81} - \frac{1}{81} + \frac{1}{6561}$$

$$= \frac{6480 - 81 + 1}{6561} = \frac{6400}{6561}$$

$$\Delta_2 = \frac{6400}{6561}$$

hence $\Delta_1 = \frac{80}{81} > 0$

$$\Delta_2 = \frac{6400}{6561} > 0$$

Thus, the time path is convergence Ans

Solved Previous papers of B.Z.U

Q. No. 4. (b) Solve the following difference equations.

i) $y_{t+2} + 3y_{t+1} - \frac{7}{4}y_t = 9$ $y(0) = 6$
 $y(1) = 3$

ii) $y_{t+2} - y_{t+1} + \frac{1}{4}y_t = 2$ $y(0) = 4$
 $y(1) = 7$